Name: _____

TOPIC TEST 2

Chapter 2: Complex numbers and De Moivre's theorem

- Time allowed: 45 minutes
- CAS calculators may be used.
- Part A: 5 multiple-choice questions (5 marks).
- Part B: 6 short-answer questions (18 marks).
- Part C: 2 analysis questions (7 marks).

Part A

Multiple-choice questions

5 multiple-choice questions
1 mark each: 5 marks
Circle the correct answer.

1 In simplest form, $4\left[\cos\left(\frac{5\pi}{6}\right)+i\sin\left(\frac{5\pi}{6}\right)\right]\div 2\left[\cos\left(\frac{2\pi}{3}\right)+i\sin\left(\frac{2\pi}{3}\right)\right]$ is

A $2\left[\cos\left(\frac{\pi}{6}\right)+i\sin\left(\frac{\pi}{6}\right)\right]$

B $2\left[\cos\left(\frac{-\pi}{6}\right)+i\sin\left(\frac{-\pi}{6}\right)\right]$

C $2\left[\cos\left(\frac{3\pi}{2}\right)+i\sin\left(\frac{3\pi}{2}\right)\right]$

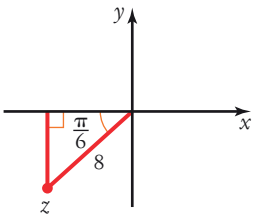
D $2\left[\cos\left(\frac{-\pi}{2}\right)+i\sin\left(\frac{-\pi}{2}\right)\right]$

E $8\left[\cos\left(\frac{\pi}{6}\right)+i\sin\left(\frac{\pi}{6}\right)\right]$

[1 mark]

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2 The complex number z is sketched below.



The value of z is:

A $4 + 4i\sqrt{3}$

B $4\sqrt{3} + 4i$

C $-4 - 4i\sqrt{3}$

D $-4\sqrt{3} - 4i$

E $-4\sqrt{3} - i\sqrt{3}$

[1 mark]

3 Expressing $(1 + i)^3$ in the form $r[\cos(\theta) + i\sin(\theta)]$ is:

A $\cos\left(\frac{3\pi}{4}\right) + i\sin\left(\frac{3\pi}{4}\right)$

B $2\left[\cos\left(\frac{3\pi}{4}\right) + i\sin\left(\frac{3\pi}{4}\right)\right]$

C $2\sqrt{2}\left[\cos\left(\frac{\pi}{4}\right) + i\sin\left(\frac{\pi}{4}\right)\right]$

D $2\sqrt{2}\left[\cos\left(\frac{3\pi}{4}\right) + i\sin\left(\frac{3\pi}{4}\right)\right]$

E $-2\sqrt{2}\left[\cos\left(\frac{\pi}{4}\right) + i\sin\left(\frac{\pi}{4}\right)\right]$

[1 mark]

4 Simplify $\left[\cos\left(\frac{\pi}{3}\right) + i\sin\left(\frac{\pi}{3}\right)\right]\left[\cos\left(\frac{\pi}{3}\right) - i\sin\left(\frac{\pi}{3}\right)\right]$:

A -1

B 1

C $-i$

D i

E 0

[1 mark]

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5 In simplest form $\dfrac{1}{\left[\cos\left(\frac{4\pi}{5}\right)+i\sin\left(\frac{4\pi}{5}\right)\right]^9}$ is:

A $\cos\left(\frac{\pi}{5}\right)+i\sin\left(\frac{\pi}{5}\right)$

B $\cos\left(\frac{-4\pi}{5}\right)+i\sin\left(\frac{-4\pi}{5}\right)$

C $\cos\left(\frac{4\pi}{5}\right)+i\sin\left(\frac{4\pi}{5}\right)$

D $\cos\left(\frac{-\pi}{5}\right)+i\sin\left(\frac{-\pi}{5}\right)$

E $\cos\left(\frac{\pi}{5}\right)-i\sin\left(\frac{\pi}{5}\right)$

[1 mark]

Part B

Short-answer questions

6 short-answer questions
18 marks
Show your working where appropriate.

1 Find $|z|$ if $z=-\frac{3}{2}-\frac{3\sqrt{3}}{2}i$.

[2 marks]

2 If $u=1-i$ and $w=-\sqrt{3}+i$ find u^2w in polar form.

[4 marks]

3 Solve the equation $z^2-3z+9=0$.

[3 marks]

4 Express $\dfrac{\sqrt{2}-i\sqrt{2}}{2}$ in mod-arg form. Hence find $\left(\dfrac{\sqrt{2}-i\sqrt{2}}{2}\right)^{-5}$ in Cartesian form.

[3 marks]

5 Show that $\left(-\sqrt{3}-i\right)^{-9}$ is purely imaginary.

[3 marks]

6 Prove for complex z that $z\overline{z}=|z|^2$.

[3 marks]

Part C

Analysis questions

2 analysis questions

11 marks

Show your working where appropriate.

1 Use the binomial theorem to expand $\left[\cos(\theta) + i\sin(\theta)\right]^3$. Hence derive a formula for $\cos(3\theta)$ in terms of $\cos(\theta)$.

[3 marks]

2 If $z = \cos(\theta) + i\sin(\theta)$ prove that $\cos(n\theta) = \frac{1}{2}\left(z^n + \frac{1}{z^n}\right)$. Derive a similar result for $\sin(n\theta)$.

[4 marks]

[Total marks: 30]

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Answers

Part A – Multiple-choice questions [1 mark each]

1 $4\left[\cos\left(\frac{5\pi}{6}\right)+i\sin\left(\frac{5\pi}{6}\right)\right]\div 2\left[\cos\left(\frac{2\pi}{3}\right)+i\sin\left(\frac{2\pi}{3}\right)\right]$

$=\frac{4}{2}\left[\cos\left(\frac{5\pi}{6}-\frac{2\pi}{3}\right)+i\sin\left(\frac{5\pi}{6}-\frac{2\pi}{3}\right)\right]$

$=2\left[\cos\left(\frac{\pi}{6}\right)+i\sin\left(\frac{\pi}{6}\right)\right]$

$\therefore$ **A** [1 mark]

2 $\sin\left(\frac{\pi}{6}\right)=\frac{y}{8}\Rightarrow y=4,\ x=4\sqrt{3},$

3rd quadrant $\Rightarrow -4\sqrt{3}-4i$

$\therefore$ **D** [1 mark]

3 $r=\sqrt{1+1}=\sqrt{2},\ \theta=\frac{\pi}{4}$

$\left[\sqrt{2}\left(\cos+i\sin\left(\frac{\pi}{4}\right)\right)\right]^3=2\sqrt{2}\left(\cos+i\sin\left(\frac{3\pi}{4}\right)\right)$

$\therefore$ **D** [1 mark]

4 $\left[\cos\left(\frac{\pi}{3}\right)+i\sin\left(\frac{\pi}{3}\right)\right]\left[\cos\left(\frac{\pi}{3}\right)-i\sin\left(\frac{\pi}{3}\right)\right]$

$=\cos^2\left(\frac{\pi}{3}\right)-i^2\sin^2\left(\frac{\pi}{3}\right)$

$=\cos^2\left(\frac{\pi}{3}\right)+\sin^2\left(\frac{\pi}{3}\right)=1$

$\therefore$ **B** [1 mark]

5 $\frac{1}{\left[\cos\left(\frac{4\pi}{5}\right)+i\sin\left(\frac{4\pi}{5}\right)\right]^9}$

$=\frac{1}{\cos\left(\frac{36\pi}{5}\right)+i\sin\left(\frac{36\pi}{5}\right)}$

$=\frac{1}{\cos\left(\frac{-4\pi}{5}\right)+i\sin\left(\frac{-4\pi}{5}\right)}$

$=\cos+i\sin\left(\frac{4\pi}{5}\right)$

$\therefore$ **C** [1 mark]

Part B – Short-answer questions

1 $|z| = \sqrt{\frac{9}{4} + \frac{27}{4}}$ **[1 mark]**

$= \sqrt{9} = 3$ **[1 mark]**

[2 marks altogether]

2 $u = 1 - i = \sqrt{2}\ \text{cis}\left(\frac{-\pi}{4}\right)$ **[1 mark]**

$w = -\sqrt{3} + i = 2\ \text{cis}\left(\frac{5\pi}{6}\right)$ **[1 mark]**

$u^2w = 2\ \text{cis}\left(\frac{-\pi}{2}\right) \times 2\ \text{cis}\left(\frac{5\pi}{6}\right)$ **[1 mark]**

$= 4\ \text{cis}\left(\frac{\pi}{3}\right)$ **[1 mark]**

[4 marks altogether]

3 $z = \frac{3 \pm \sqrt{(-3)^2 - 4(1)(9)}}{2}$ **[1 mark]**

$= \frac{3 \pm \sqrt{-27}}{2}$ **[1 mark]**

$= \frac{3 \pm 3i\sqrt{3}}{2}$ **[1 mark]**

[3 marks altogether]

4 $\frac{\sqrt{2} - i\sqrt{2}}{2} = \text{cis}\left(\frac{-\pi}{4}\right)$ **[1 mark]**

$\left[\text{cis}\left(\frac{-\pi}{4}\right)\right]^{-5} = \text{cis}\left(\frac{5\pi}{4}\right)$ **[1 mark]**

$= \text{cis}\left(\frac{-3\pi}{4}\right) = -\frac{1}{\sqrt{2}} - \frac{i}{\sqrt{2}}$ **[1 mark]**

[3 marks altogether]

5 $\left(-\sqrt{3} - i\right)^{-9} = \left[2\text{cis}\left(\frac{-5\pi}{6}\right)\right]^{-9}$ **[1 mark]**

$= \frac{1}{512}\ \text{cis}\left(\frac{45\pi}{6}\right) = \frac{1}{512}\ \text{cis}\left(\frac{-\pi}{2}\right)$ **[1 mark]**

$\frac{1}{512}\left[\cos\left(\frac{-\pi}{2}\right) + i\sin\left(\frac{-\pi}{2}\right)\right] = \frac{-i}{512}$, which is purely imaginary. **[1 mark]**

[3 marks altogether]

6 Let $z = x + yi$ where x and y are real.

$z\bar{z} = (x + yi)(x - yi) = x^2 + y^2$ **[1 mark]**

$|z|^2 = |x + yi|^2 = x^2 + y^2$ **[1 mark]**

$\therefore z\bar{z} = |z|^2$ QED **[1 mark]**

[3 marks altogether]

Part C – Analysis questions

1 $\cos(3\theta) + i\sin(3\theta)$

$= \cos^3(\theta) + 3i\cos^2(\theta)\sin(\theta) - 3\cos(\theta)\sin^2(\theta) - i\sin^3(\theta)$ **[1 mark]**

Equating real parts: $\cos(3\theta) = \cos^3(\theta) - 3\cos(\theta)\sin^2(\theta)$

$= \cos^3(\theta) - 3\cos(\theta)\left[1 - \cos^2(\theta)\right]$ using $\sin^2(\theta) = \left[1 - \cos^2(\theta)\right]$ **[1 mark]**

$= \cos^3(\theta) - 3\cos(\theta) + 3\cos^3(\theta) = 4\cos^3(\theta) - 3\cos(\theta)$ **[1 mark]**

[3 marks altogether]

2 $z^n = \cos(n\theta) + i\sin(n\theta)$, $\dfrac{1}{z^n} = z^{-n} = \cos(-n\theta) + i\sin(-n\theta) = \cos(n\theta) - i\sin(n\theta)$ **[1 mark]**

then adding $z^n + \dfrac{1}{z^n} = 2\cos(n\theta)$ **[1 mark]**

so $\cos(n\theta) = \dfrac{1}{2}\left(z^n + \dfrac{1}{z^n}\right)$ **[1 mark]**

Similarly, subtracting $z^n - \dfrac{1}{z^n} = 2i\sin(n\theta)$

so $\sin(n\theta) = \dfrac{1}{2i}\left(z^n - \dfrac{1}{z^n}\right)$ **[1 mark]**

[4 marks altogether]

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